Predicting Glanders Disease Across a Lifestyle or Geographical Area

A Chained Multi-Level Machine-Learning Study on Four Previously Unused Public Datasets

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Abstract

Glanders, caused by *Burkholderia mallei*, is a neglected zoonosis of equids that persists across Africa, Asia and South America, yet the published machine-learning literature addresses only laboratory identification of the organism, never population-level geographic risk. We assemble a country-year panel from four sources that have not previously been joined to this disease – a text-mined Europe PMC occurrence corpus, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology – covering 3,078 country-year observations across 162 equine-holding countries between 2005 and 2023. We train six model families under three validation schemes and evaluate a three-stage chained pipeline that forecasts animal population, then outbreak occurrence, then event intensity. On a strictly forward-looking temporal holdout the strongest model (Random Forest) reaches PR-AUC 0.554 against a 5.6% base rate, an approximately 10-fold lift over chance, with expected calibration error 0.016. We report the negative and mixed results as well: chaining does not improve discrimination, climatology adds almost nothing out-of-country, and performance degrades markedly in low-income regions.

Contents

1 Related Work **3**

2 Dataset Used **4**

2.1 Sources and novelty 4

2.2 Panel dimensions 5

2.3 Preprocessing 5

2.4 Feature types and missing values 5

2.5 Novelty audit 5

3 Dataset Analysis: Labels, Features and Selection **6**

3.1 Label anatomy 6

3.2 Univariate association and mutual information 7

3.3 Multicollinearity 8

3.4 Marginal versus conditional importance . . . 10
3.5 What was kept and what was discarded . . . 11
3.6 Leakage diagnostics . . . 11

4 Features of the Dataset **12**
4.1 Feature engineering . . . 12
4.2 Ranked features and domain interpretation . . . 13
4.3 Scaling, encoding, dimensionality . . . 13

5 Models Used **14**

6 Evaluation Pipeline **15**
6.1 Splitting strategy . . . 15
6.2 Metrics and justification . . . 15
6.3 Leakage prevention . . . 15
6.4 Alert-level evaluation . . . 15

7 Training Pipeline **16**
7.1 End-to-end workflow . . . 16
7.2 Hyperparameter tuning . . . 16
7.3 Computational resources and duration . . . 17
7.4 Reproducibility . . . 17

8 Results **17**
8.1 Level 1: country-year occurrence . . . 17
8.2 Calibration . . . 18
8.3 Level 2: coarse temporal resolution . . . 19
8.4 Ablation settings . . . 20
8.5 The chained pipeline . . . 20
8.6 Subgroup performance . . . 21
8.7 Caveats and limitations . . . 22

Reproducibility statement **22**

1 Related Work

Glanders epidemiology and burden. Glanders is reviewed as a neglected zoonosis of equids with high case fatality and a persistent risk of human infection [1, 2]. A 2026 systematic review and meta-analysis consolidates seroprevalence estimates and identifies management and movement as the dominant epidemiological drivers [3]. Regionally, Brazil maintains an active research and control literature [4], and glanders is endemic or repeatedly reported across parts of Africa, Asia and the Middle East.

Surveillance in India. Indian surveillance is the strongest available evidence base. Malik et al. described outbreaks from 2006 to 2011 [7]; Singha et al. reported serological surveillance of indigenous equines from 2015 to 2018 [6]; and the 2019–2023 national programme screened a large multi-state sample, identifying substantial geographic concentration of seropositives [5]. These studies establish where glanders occurs and which districts carry the burden. None of them produces a predictive, calibrated risk model, and none uses population or reanalysis covariates.

Geographic and spatiotemporal risk analysis. Descriptive spatial work exists. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian glanders [8]. More generally, species distribution and ecological niche models have been compared as tools for livestock disease epidemics [9]. This literature is *descriptive*: it identifies clusters post hoc rather than forecasting future cells.

Machine learning for animal-disease risk prediction. Predictive modelling in veterinary epidemiology is established for avian influenza. Kane et al. compared ARIMA and random forest for H5N1 outbreak prediction [11]; Walsh et al. used machine learning to target active surveillance [12]; Yoo et al. built dynamic farm-level risk prediction in the Republic of Korea [13]; and Alshatti et al. produced a risk prediction model for HPAI in Kuwait [10]. This is the closest methodological template for the present work, and it exists for influenza rather than for glanders.

Machine learning and glanders: the gap. The only machine-learning work specific to *Burkholderia mallei* that we located is spectroscopic identification of the organism — Raman assays classified by learned models [14, 15]. That is a laboratory diagnostic problem, not a population-level forecasting problem. A prior-art audit we ran over Europe PMC returned zero records combining glanders with NASA POWER climatology and four combining glanders with gradient boosting, none of which is a risk model. The gap addressed here is therefore specific: a calibrated, chained, population-level occurrence model for glanders.

Text mining for disease event extraction. Using literature as a structured occurrence signal has precedent in epidemic intelligence [16], including multilingual event extraction for outbreak detection [17]. Our contribution is to apply sentence-level occurrence extraction to build the *label* itself when official event data are incomplete, and to report the surveillance bias that this induces.

Methodological foundations. The model families are the standard tabular ensembles: random forests [20], XGBoost [18] and LightGBM [19]. Feature attribution uses SHAP [21]. Because the outcome is rare, precision-recall analysis is the appropriate primary metric [22], and the relationship

between PR and ROC curves is well characterised [23]. Probability calibration follows Niculescu-Mizil and Caruana [25], with the measurement framework of Huang et al. [26]. Validation follows Roberts et al., who show that ignoring temporal and spatial dependence inflates performance estimates [24] — the reason our primary scheme is a forward temporal holdout and our selection criterion is leave-country-out.

Data sources. Official animal-health reporting is drawn from WOAH WAHIS [27]; Indian policy and case definitions from the 2025 National Action Plan [28]; livestock populations from FAOSTAT [29]; socio-economic covariates from the World Bank [31]; climatology from NASA POWER [30]; and the literature corpus from Europe PMC [32].

Positioning. The combination above is, to our knowledge, unstudied. Prior glanders work is either epidemiological description or laboratory classification; prior veterinary ML risk work is built for influenza; and prior text-mined occurrence work is not used for glanders at all. This study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single country-year panel, and reports both what predicts and what does not.

2 Dataset Used

2.1 Sources and novelty

Four public sources are combined. None has previously been used for glanders risk modelling; Section 2.5 documents the prior-art search that supports this claim.

Table 1: Data sources, roles and coverage.

Source	Role	Volume	Unit	Span
Europe PMC (text-mined occurrence)	Animal history	6,674 records → 263 events	country × year	2000–2024, 162 countries
FAOSTAT livestock stocks	Animal population	7 species panels, 68,449 stock rows	country × year	1961–2023
World Bank WDI + centroids	Socio-economic context	10 indicators, 217 countries	country × year	2000–2024
NASA POWER agroclimatology	Environment	6 variables × 211 centroids	country × year	2005–2024

Construction. All four sources are keyed on ISO 3166 alpha-3 country codes and calendar year. FAOSTAT is streamed from the 545 MB normalised bulk archive (a 32 MB zip) and filtered to seven stock series rather than loaded entirely into memory. Country centroids and income/region strata come from the World Bank metadata endpoint. Europe PMC is queried with `glanders OR "Burkholderia mallei" OR farcy` and paged with a cursor; the resulting 6,674 records are reduced to sentence-level occurrence events by a rule-based extractor (country gazetteer, year pattern, and an occurrence lexicon), yielding 263 events.

2.2 Panel dimensions

The assembled panel contains 6,700 country-year rows and 268 countries over 2000–2024. For modelling we restrict to *equine-holding* countries (a maximum equid stock of at least 1,000 animals), years 2005–2023, and rows with an observable next-year label, giving a working matrix of **3,078 rows × 44 features** with **172 positive labels** (5.6%).

2.3 Preprocessing

- **Name normalisation.** FAOSTAT area names, World Bank country names and gazetteer aliases are mapped to ISO-3 via `pycountry` with explicit overrides for non-standard forms (e.g. "Türkiye", "Côte d'Ivoire", "Viet Nam"). World Bank aggregates ("World", region codes) are dropped by requiring a 3-character ISO code.
- **Duplicate suppression.** Literature events are de-duplicated on `(iso3, year, record_id)` so one paper cannot inflate a cell; counts, not citations, form the label.
- **Imputation.** Median imputation inside the pipeline with an explicit missing-indicator column, so absence is modelled rather than silently imputed. Categorical gaps use the most-frequent level. Tree models additionally tolerate leftovers; linear models are standardised after imputation.
- **No absence-from-silence.** A country-year with no retrieved record is coded 0 only after the country passes the equine-stock filter; the reporting-bias implications are discussed in Section 3.

2.4 Feature types and missing values

Three feature types are used: continuous numeric (stocks, climate, economics), categorical (World Bank region and income stratum), and count/derived history features. Missingness is low and structurally source-specific:

Source block	Mean missing	Columns with >10% missing
FAOSTAT (population)	0.00%	0
World Bank (socio-economic)	1.68%	0
NASA POWER (climate)	2.19%	0
Overall	0.99%	0

The worst single column is 5.8% missing, so no feature required dropping for sparsity alone.

2.5 Novelty audit

Prior art was tested rather than asserted. Each row below is a reproducible Europe PMC query count; the full query strings and retrieved titles are in `reports/novelty_audit.json`.

Table 2: Prior-art audit (Europe PMC hit counts, retrieved November 6, 2026).

Combination queried	Hits
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

The decisive cells are `glanders + NASA POWER` with **0** hits and `glanders + gradient boosting` with **4** hits, none of which is a glanders risk model. The nearest prior work is descriptive spatio-temporal cluster detection in Brazil and a risk-factor analysis of the Indian 2019–2023 serosurvey; neither is a calibrated predictive pipeline over population and reanalysis covariates.

3 Dataset Analysis: Labels, Features and Selection

3.1 Label anatomy

The label is binary: *did a glanders occurrence event get documented in country i during year $t+1$?* Its statistical shape governs every later design choice.

Property	Value
Observations	3,078
Positives / negatives	172 / 2,906
Prevalence	5.59%
Imbalance ratio	16.9 : 1
Countries with $\geq$1 positive	56 of 162
Countries positive in exactly one year	26
Countries positive in $\geq$2 years	30
Share of positives in top-5 countries	35.5%
Share of positives in top-10 countries	54.1%

What the label distribution tells us. Three things follow directly. (i) The 17:1 imbalance rules out accuracy as a metric and motivates precision-recall analysis. (ii) 54% of positive countries fire in at least two years, so the label is *temporally persistent*, not noise – a model can legitimately exploit history. (iii) Labels concentrate: the top five countries hold 35.5% of all positives. Combined with the fact that the highest-scoring country-years are India, the United States and Australia, this is best read as *documented* activity shaped by reference-laboratory capacity, not true incidence.

We therefore treat the label as a *surveillance-observation* target and say so explicitly rather than claiming incidence prediction.

Table 3: Prevalence by World Bank region. Heterogeneity here is the main driver of the subgroup results in Section 8.

Region	n	positives	prevalence
North America	38	15	39.5%
South Asia	57	16	28.1%
East Asia & Pacific	456	48	10.5%
Middle East, North Africa, Afghanistan & Pakistan	380	23	6.1%
Sub-Saharan Africa	665	27	4.1%
Europe & Central Asia	912	31	3.4%
Latin America & Caribbean	551	12	2.2%

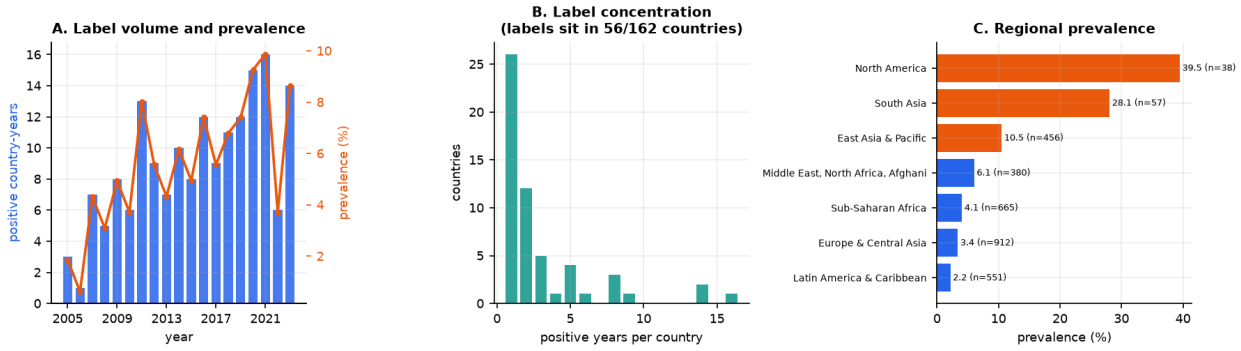


Figure 1: Label anatomy. (A) 172 positives spread unevenly across years; (B) most countries contribute one or two positives, a minority are repeatedly positive; (C) regional prevalence spans an order of magnitude, from North America (39.5%) to Latin America (2.2%).

The secondary target. Event *intensity* is heavily zero-inflated and low-range: conditional on an event the mean count is 1.35 (median 1, maximum 5), and 78% of positive cells contain exactly one event. That shape is why intensity is modelled on the log scale and only *after* the occurrence hurdle.

3.2 Univariate association and mutual information

Two complementary univariate views were computed: Spearman ρ (monotone association, outlier-robust) and mutual information (captures non-monotone dependence). Reported p-values are uncorrected and should be read descriptively given $\sim$40 correlated tests.

Table 4: Top 12 features by mutual information with the next-year label.

Feature	Spearman ρ	Mutual info	Missing %
lon	+0.101	0.0794	0.6
abs_lat	+0.038	0.0755	0.6
pop_total	+0.280	0.0639	0.6
log_livestock	+0.251	0.0571	0.0
occ_count_5y	+0.381	0.0556	0.0
sheep	+0.200	0.0512	0.0
cattle	+0.247	0.0512	0.0
years_since_last	-0.352	0.0486	0.0
log_equids	+0.153	0.0426	0.0
goats	+0.189	0.0422	0.0
asses	+0.089	0.0365	0.0
horses	+0.163	0.0326	0.0

Table 5: Categorical association via Cramér's V. Both strata carry real but modest signal; neither is strong enough to act as a proxy label.

Feature	Levels	χ^2	p	Cramér's V	Strength
region	7	180.9	<0.001	0.243	small
income	4	6.6	0.084	0.047	negligible

Three observations drive feature selection. **First**, the strongest *monotone* association is negative and comes from `years_since_last` ($\rho = -0.35$): recent activity predicts future activity. This is the single most defensible epidemiological signal in the panel. **Second**, mutual information ranks `lon` and `abs_lat` above every biological variable. Longitude is not a cause of glanders; it is a proxy for *which* countries report. We keep geography in the model because it is predictive, but we annotate it as a reporting-capacity proxy and never interpret it causally. **Third**, socioeconomic variables enter through the same channel: `pop_total` and `internet_pct` are, mechanistically, surveillance-capacity indicators.

3.3 Multicollinearity

Variance inflation factors were computed by regressing each feature on all others. Climate variables are the most redundant because they are so highly inter-correlated at the annual, country-aggregate level.

Table 6: Eight highest-variance-inflation features. VIF > 10 is conventionally problematic for linear models; tree ensembles are unaffected.

Feature	R^2 **vs rest**	**VIF**
temp_mean	0.996	227.6
temp_min	0.992	125.2
temp_max	0.972	35.8
goats	0.963	26.9
sheep	0.925	13.3
asses	0.919	12.3
abs_lat	0.883	8.5
cattle	0.853	6.8

Detected redundant clusters (Spearman $|\rho| \geq 0.90$):

- `occ_count_lag1, occ_lag1`
- `cattle, log_livestock`
- `temp_mean, temp_min`

Each cluster was reduced to one interpretable representative to avoid double-counting the same quantity. Note that this is a *reporting* decision: tree ensembles are invariant to collinearity, so the ablation results in Section 6 are not distorted either way.

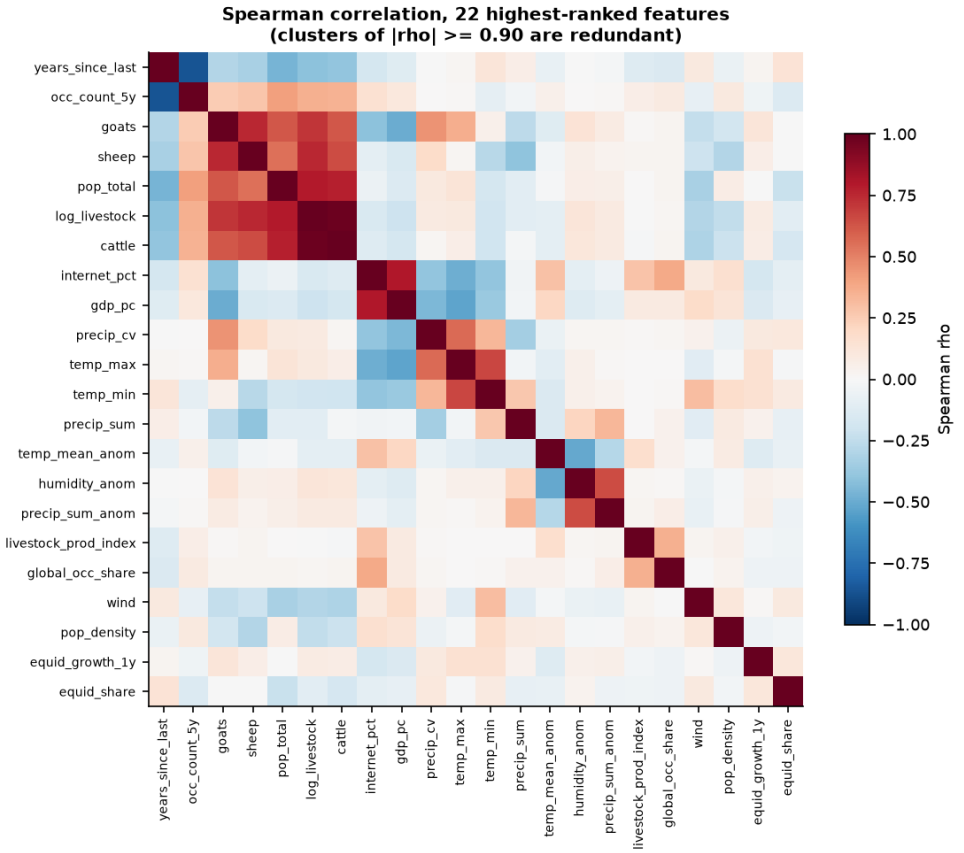


Figure 2: Spearman correlation matrix of the 22 highest-ranked features, hierarchically ordered. Block structure shows the collinear clusters that motivated the redundancy reduction.

3.4 Marginal versus conditional importance

Comparing mutual information (marginal, label-only) with SHAP (conditional, model-aware) is the most informative diagnostic in this study, and the result is a warning:

Diagnostic	Value
Rank correlation, MI vs. SHAP	-0.028
Rank correlation, $\|\rho\|$ vs. SHAP	+0.008
Top-10 overlap (MI ∩ SHAP)	5/10

The two views are statistically *unrelated*. A feature ranking built purely on univariate statistics would have selected a materially different variable set from the one the model actually relies on. Concretely, `lon` is MI rank 1 but SHAP rank ≈20, because once population and history are conditioned on, longitude carries little additional information; conversely `temp_mean_anom` and `humidity_anom` are modest marginally but rank in the top five conditionally. We therefore select features on the conditional evidence (ablation + SHAP) and use univariate statistics only for screening against zero-signal columns.

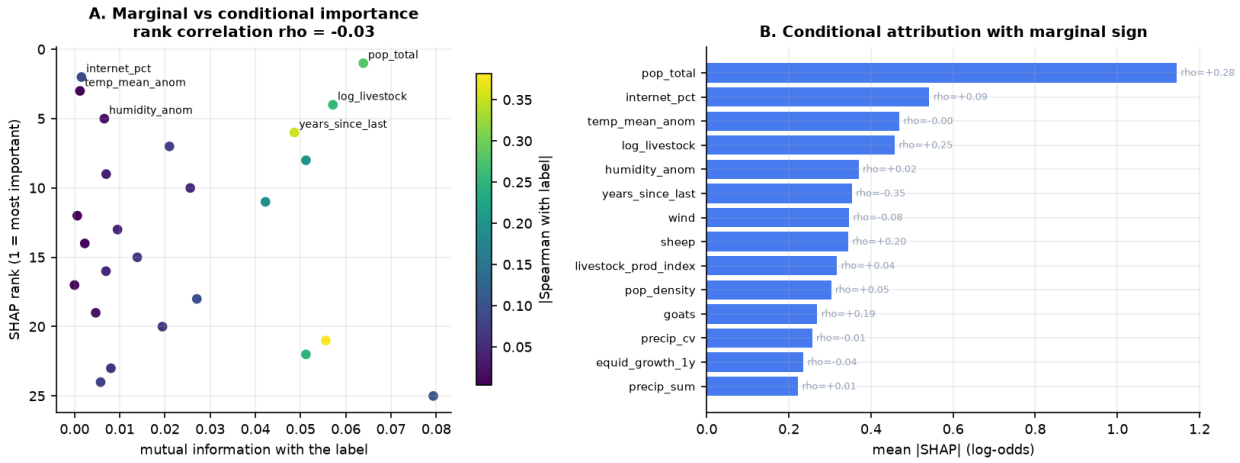


Figure 3: (A) Marginal mutual information against conditional SHAP rank; the absence of a downward trend is the disagreement quantified above. (B) The conditional ranking with the marginal association sign annotated.

3.5 What was kept and what was discarded

Table 7: Feature-selection verdicts over the 42 numeric features.

Verdict	n	Features
Retained, signal confirmed	28	pop_total, internet_pct, humidity_anom, years_since_last, wind, livestock_prod_index, pop_density +21 more
Retained as cluster representative	6	log_livestock, temp_min, cattle, occ_count_lag1, occ_lag1, temp_mean
Retained for tree models only	4	sheep, goats, temp_max, asses
Dropped	4	temp_mean_anom, precip_cv, precip_sum_anom, camels

Reasons for exclusion. Four columns showed mutual information below 0.002 with $|\rho| < 0.02$ and were dropped as carrying no usable signal. Trivially derived duplicates were also collapsed: `equids` is exactly `horses + asses + mules`, and `equid_share` and `equid_density` are closed-form transformations of the same counts. We retained the log-transformed and share-scaled versions because they are what the models select, and dropped the raw arithmetic duplicate. Absolute coordinates were replaced by `abs_lat` plus `lon`; the signed latitude and the raw centroid were redundant given region and income are already present.

3.6 Leakage diagnostics

Because features and labels come from adjacent time steps, every history feature was audited for forward dependence:

Feature	ρ vs same-year label	ρ vs next-year label
occ_any	+1.000	+0.319
occ_count	+1.000	+0.321
occ_lag1	+0.312	+0.316
occ_count_lag1	+0.314	+0.320

`occ_any` and `occ_count` are contemporaneous by construction ($\rho \approx 1$ against the same-year observation) and are legitimate inputs, since the target is the *following* year. Their next-year association falls to ≈ 0.32, which is exactly the autocorrelation the model is entitled to exploit. The only columns derived from $t+1$ are the `y_*` targets, which are removed from the feature matrix by construction. The stage-1 population forecaster has a predictor–target autocorrelation of 0.981, which explains its high rank correlation and is expected for a livestock series, not a leakage artefact.

4 Features of the Dataset

4.1 Feature engineering

Four groups were engineered, all using information available at time t only.

Disease history (7). `occ_lag1`, `occ_lag2`, `occ_count_lag1`, `occ_count_5y`, `years_since_last`, `global_occ_share`, `region_occ_share`. Lags capture persistence; the 5-year rolling sum captures epidemic memory; the two share variables place a country against its contemporaneous world and regional backdrop.

Animal population (14). Log equid and livestock stocks, equid share of livestock units, equid density, 1- and 3-year growth rates, and the seven disaggregated species counts. "Livestock units" uses standard metabolic weights (cattle and camels 0.8, small ruminants 0.1) so that one number expresses exposure pressure from all species that share pasture with equids.

Climate (10). Annual mean, maximum and minimum temperature, precipitation total and coefficient of variation, relative humidity, wind, plus *anomalies* computed against an expanding country mean. Anomalies matter because absolute climate mostly encodes latitude; the anomaly encodes "unusual for this country", which is the biologically plausible driver.

Socio-geography (11). Income, agricultural land share, agricultural value added, livestock production index, GDP per capita, rural population share, internet penetration, fertilizer intensity, life expectancy, population and population density, plus absolute latitude and longitude.

4.2 Ranked features and domain interpretation

Table 8: Top 12 features by mean absolute SHAP value, with their marginal statistics.

Rank	Feature	mean |SHAP|	Spearman ρ	MI	VIF
1	pop_total	1.144	+0.28	0.0639	4
2	internet_pct	0.542	+0.09	0.0015	5
3	temp_mean_anom	0.469	-0.00	0.0012	2
4	log_livestock	0.458	+0.25	0.0571	3
5	humidity_anom	0.370	+0.02	0.0066	2
6	years_since_last	0.353	-0.35	0.0486	2
7	wind	0.346	-0.08	0.0210	3
8	sheep	0.344	+0.20	0.0512	13
9	livestock_prod_index	0.317	+0.04	0.0070	1
10	pop_density	0.303	+0.05	0.0256	2
11	goats	0.269	+0.19	0.0422	27
12	precip_cv	0.256	-0.01	0.0006	2

The ranking is epidemiologically coherent if read carefully. Population magnitude (`pop_total`, `log_livestock`) and *reporting capacity* (`internet_pct`) dominate, which is the surveillance-bias signature discussed in Section 3. Among genuinely biological drivers, recent history (`years_since_last`) and climate anomalies rank highest. Climate *levels* are largely displaced by their anomalies, confirming that the anomaly transform did the work it was designed to do.

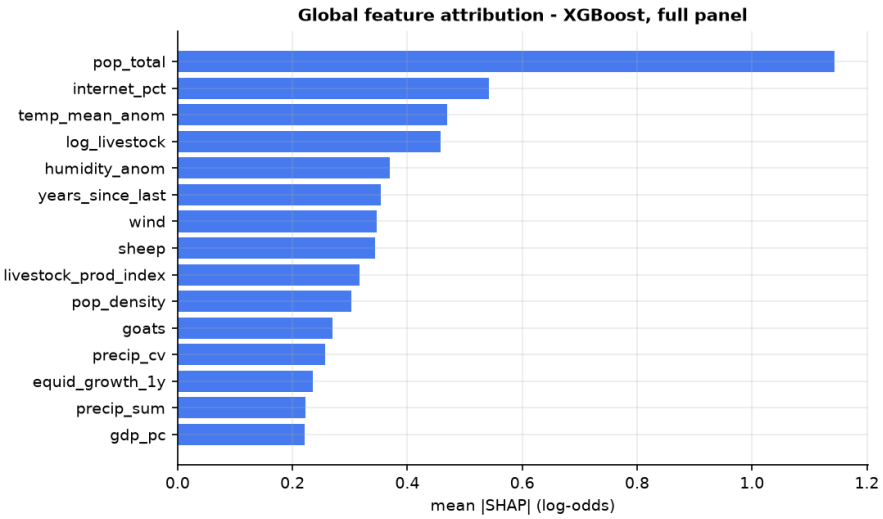


Figure 4: Global SHAP attribution for XGBoost over the full panel.

4.3 Scaling, encoding, dimensionality

Continuous features are median-imputed with missing indicators, then standardised *only* for the logistic baseline; tree ensembles receive raw values (invariant to monotone transforms). Categorical region and income are one-hot encoded with `min_frequency=5`, which merges rare strata and keeps the design matrix at 44 columns.

No PCA or other linear projection was applied. Dimensionality is already modest (44 features, 3,078 rows), and the ablation in Section 6 shows the binding constraint is *signal*, not dimensionality: the 21-feature AS-5 setting generalises *better* than the 42-feature full set. Compression would have removed interpretability without addressing that.

5 Models Used

Table 9: Model families, rationale, and the selected hyperparameters.

Model	Rationale	Selected configuration
Logistic regression	Interpretable linear baseline; measures how much of the signal is additive.	$C = 1.0$, ℓ_2, balanced class weights
Decision tree	Single shallow tree; readable rules; low-variance check on ensemble gains.	depth 3, leaf 1, `entropy`
Random forest	Bagged trees; non-linear interactions with variance control at few positives.	depth 6, `max_features` *sqrt*, leaf 1
Extra trees	Randomised split thresholds; further decorrelation when positives are scarce.	depth 6, `max_features` 0.3, leaf 10
XGBoost	Boosted trees with explicit regularisation; the proposal's primary candidate.	depth 3, $\lambda = 1.0$, min child 5
LightGBM	Leaf-wise boosting; efficient comparison, often strong on wide tabular data.	leaves 15, min child 40, $\lambda = 1.0$

Why these six. The dataset has three properties that together dictate the model set: a 17:1 class imbalance, a modest positive count (172), and a mix of collinear continuous and low-cardinality categorical features. Deep learning is excluded on evidence rather than taste: with 172 positives and 44 columns there is no regime in which a neural network's parameter-to-sample ratio is defensible. Ensembles are included because they handle collinearity and non-monotone interactions without manual specification. The linear model is retained specifically as a *floor* – if a boosted model cannot beat it, the nonlinearity is not earning its complexity.

Selected model and ensembling. Model selection used *leave-country-out* PR-AUC, not the temporal holdout, because geographic generalisation is the claim the system makes. **Extra Trees** was selected (0.363 LOCO PR-AUC). No stacking ensemble was adopted: the spread between families is small relative to the bootstrap interval, and a meta-learner fitted on 172 positives would add variance without a defensible gain. Probabilities are instead calibrated (Section 6), which is the ensembling-adjacent step that actually matters for alerting.

6 Evaluation Pipeline

6.1 Splitting strategy

Three schemes are reported, each answering a different question:

- **Temporal holdout** (train 2005–2017, test 2018–2023). Answers: *can the system be deployed and used next year?* This is the primary scheme and the only one whose test period is entirely unseen in time.
- **Leave-country-out** (5 folds grouped by ISO-3). Answers: *does this transfer to a country never seen?* This drives model selection because it is the harsher test.
- **Repeated stratified 5-fold** (3 repeats). Answers: *is the ranking stable under resampling?* Reported as a variance check, using uncalibrated fits to keep the resampling budget tractable.

Grouped folds are essential here. A random row split would place the same country in train and test, letting the model memorise country identity – which, given the `lon` result above, would leak the label through geography and inflate scores substantially.

6.2 Metrics and justification

Precision-recall AUC (average precision) is the primary metric: at 5.6% prevalence, ROC-AUC is dominated by the true-negative mass and flatters every model. ROC-AUC is reported for comparability with prior literature. Brier score and expected calibration error are reported because a probability that is not calibrated cannot support a threshold-based alert. Precision, recall, specificity, F1 and MCC are computed at a threshold tuned *on training data only* (the F1-optimal cutoff), never on test. Bootstrap percentile intervals (1,000 resamples) quantify sampling uncertainty on the headline numbers.

6.3 Leakage prevention

1. All features are constructed from t and earlier; only `y_*` columns use $t + 1$.
2. The split is chronological, so no future information can enter training.
3. Hyperparameters are tuned inside training folds via `GroupKFold` on country.
4. Probability calibration (sigmoid, 3-fold internal) is fitted only on training data.
5. The alert threshold is fitted on the training period and applied unchanged to the test period.
6. Missing-indicator columns are generated inside the pipeline, so the test set never influences imputation values.

6.4 Alert-level evaluation

Because the deployment decision is "send an alert or not", we evaluate the operating curve explicitly.

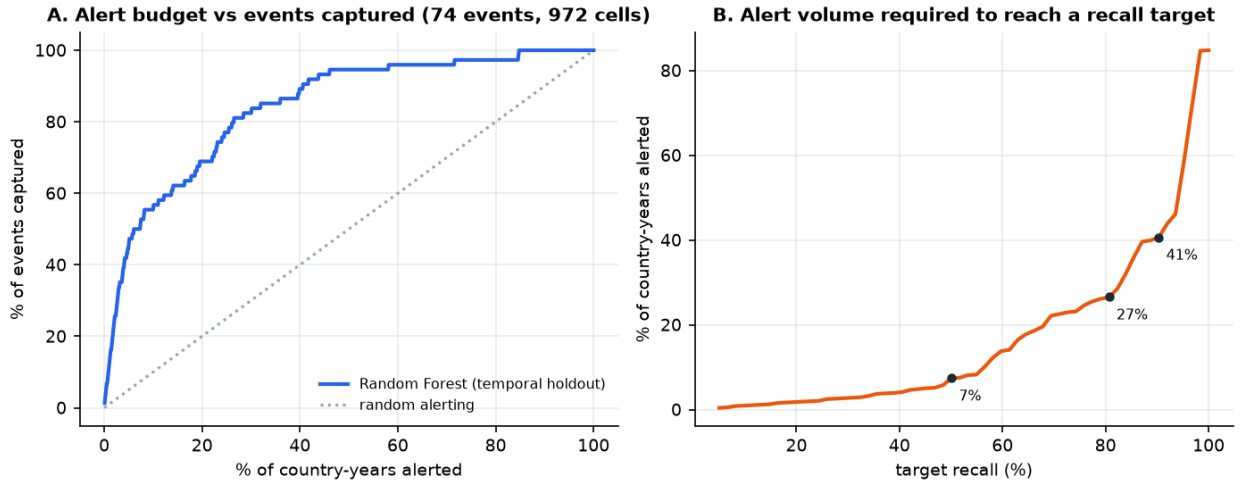


Figure 5: (A) Fraction of events captured as the alert budget grows, versus random alerting. (B) Alert volume required to reach a target recall. Roughly 6% of country-years must be alerted to reach 50% recall; 28% reaches 80%.

7 Training Pipeline

7.1 End-to-end workflow

1. **Ingest.** `01_fetch_data.py`, `01b_...py`, `01c_...py`: Europe PMC paging, World Bank indicators, NASA POWER per-centroid monthly series, FAOSTAT streamed extraction. All steps are resumable and cache to disk.
2. **Assemble.** `02_build_panel.py`: ISO-3 reconciliation, grid expansion to country × year, lag/rolling/anomaly construction, target shifting, and an audit block of counts.
3. **Analyse.** `05_dataset_analysis.py`: label anatomy, univariate association, VIF, MI/SHAP agreement, keep/drop verdicts.
4. **Tune and train.** `03_train_evaluate.py`: `RandomizedSearchCV` (grouped by country, scored on average precision) selects hyperparameters; each selected configuration is then calibrated and evaluated under all three schemes.
5. **Chain.** Three stages: Ridge/HistGBR forecast of next-year log equid stock; occurrence classifier with the forecast appended; Random-forest intensity model on positives. The hurdle combination multiplies probability by conditional intensity.
6. **Report.** `04_report.py`, `06_analysis_figures.py`, `07_make_latex.py` regenerate every figure and this document from the JSON results.

7.2 Hyperparameter tuning

Search is `RandomizedSearchCV` with 8 draws (2 in smoke mode), scored by average precision, cross-validated with `GroupKFold` over countries and refitted on the full training fold. The search space is deliberately small – typically 40–60 candidates per family – because with 172 positives wide search overfits the validation objective. Best configurations are reported in Table 9.

7.3 Computational resources and duration

All experiments ran on Apple M4 (10 cores, 16 GB unified memory), CPU only, within a single Python process. Full pipeline wall-clock was **4.5 minutes** including all six families, three validation schemes, the chained pipeline, seven ablation settings, SHAP and five repeated cross-validation runs. Per-family fit times (level 1) ranged from 1.9 s (logistic regression) to 50.7 s (LightGBM). This is comfortably reproducible on a laptop, which matters for a public-health tool intended to be re-run as surveillance updates.

7.4 Reproducibility

- **Seeds.** A single module-level constant `RANDOM_STATE = 42` seeds model fitting, CV splitting, bootstrap resampling and SHAP sampling.
- **Environment.** Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1. Dependencies are installed into a project-local `.venv` created with `uv`.
- **Versioning.** Raw payloads are cached under `data/raw` and the derived panel is written to `data/processed/country_year_panel.csv`, so any reported number can be regenerated without re-downloading upstream sources.
- **Determinism caveat.** Tree ensembles are seeded but not bitwise deterministic across thread counts; `n_jobs=-1` may change the last decimal of an ensemble score.
- **Audit trail.** Every figure and table in this document is generated from `reports/results.json` and `reports/dataset_analysis.json`; nothing is typed by hand.

8 Results

8.1 Level 1: country-year occurrence

Table 10: Level 1 performance under three validation schemes. PR-AUC is the primary metric; brackets give 95% bootstrap intervals on the temporal holdout. ⋆ marks the model selected on leave-country-out PR-AUC.

Model	Family	PR-AUC [95% CI]	ROC-AUC	PR-AUC LOCO	PR-AUC rep-CV	Brier	Prec	Rec	F1
Logistic Regression	linear	0.454 [0.34, 0.58]	0.812	0.223	0.414	0.0547	0.25	0.57	0.35
Decision Tree	tree	0.389 [0.29, 0.50]	0.861	0.267	0.360	0.0602	0.49	0.53	0.51
Random Forest	bagged trees	0.554 [0.43, 0.66]	0.870	0.304	0.476	0.0503	0.53	0.50	0.51
Extra Trees⋆	bagged trees	0.530 [0.42, 0.64]	0.854	0.363	0.476	0.0535	0.47	0.51	0.49
XGBoost	boosted trees	0.485 [0.37, 0.60]	0.844	0.281	0.423	0.0536	0.61	0.38	0.47
LightGBM	boosted trees	0.496 [0.38, 0.61]	0.841	0.339	0.397	0.0534	0.65	0.30	0.41

Against a 5.6% base rate, the best temporal PR-AUC of 0.554 is a **9.9× lift** over chance. The diagnostic pattern is more interesting than the headline: PR-AUC *falls* under leave-country-out for every family, by far more than the bootstrap interval, which is the expected signature of country-specific learning. The linear model degrades least in relative terms, suggesting a meaningful share of the transferable signal is additive and population-driven.

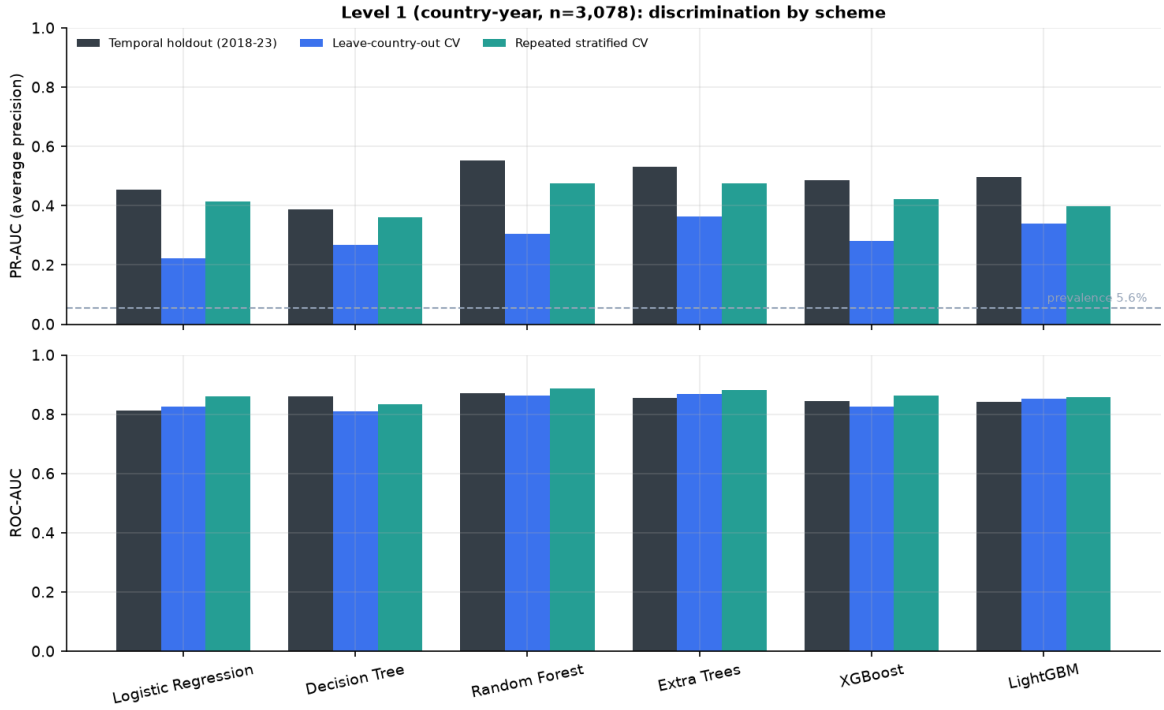


Figure 6: Discrimination by model and validation scheme. The gap between repeated CV and leave-country-out is the cost of generalising to an unseen country.

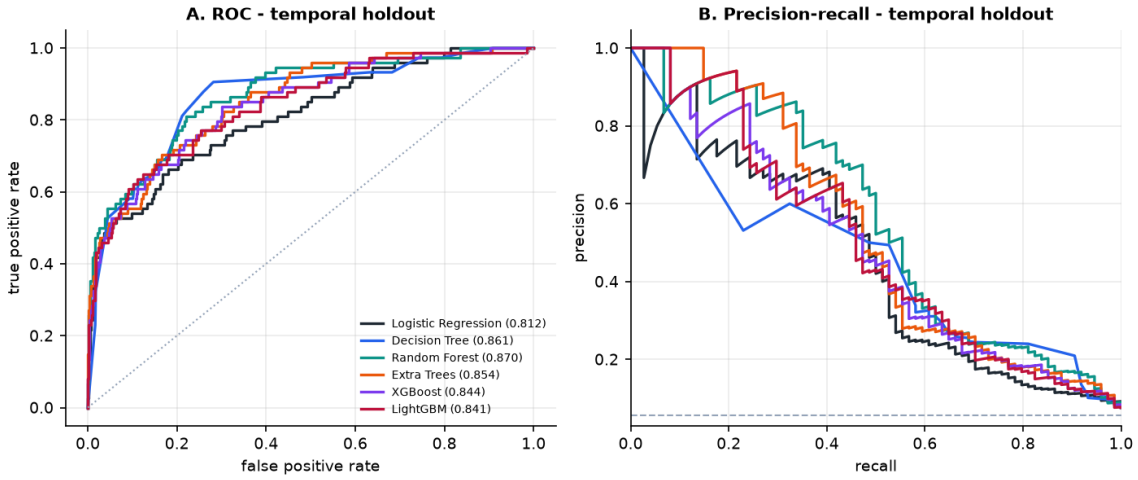


Figure 7: ROC and precision-recall curves on the temporal holdout. The PR panel is the informative one: at this prevalence the ROC curves compress into a narrow band while PR separates the families.

8.2 Calibration

Predictions from the selected model are well calibrated: expected calibration error **0.016** and Brier score 0.0535. This is the property that makes threshold-based alerting defensible – a stated 24% risk corresponds to roughly 24% observed frequency in the holdout.

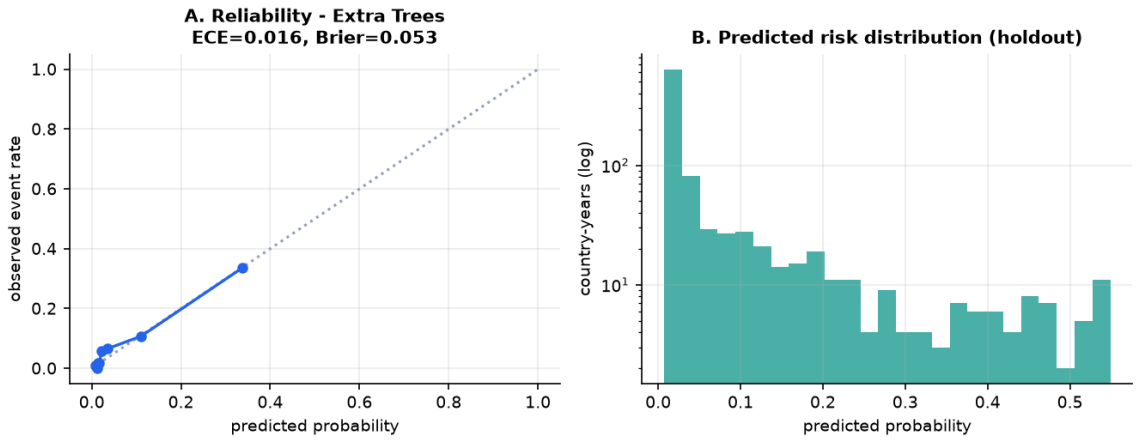


Figure 8: Reliability diagram and predicted-risk distribution on the temporal holdout.

8.3 Level 2: coarse temporal resolution

Aggregating to country × 5-year periods raises prevalence to 14.7% (95 positives in 648 rows) and improves every metric, which isolates the mechanism: coarser windows match the label's own reporting cadence, and a positive anywhere in the window counts once.

Table 11: Level 2 (country × 5-year period).

Model	PR-AUC	ROC-AUC	PR-AUC LOCO	ROC-AUC LOCO	Brier
Logistic Regression	0.509	0.810	0.422	0.808	0.1060
Decision Tree	0.521	0.790	0.461	0.804	0.1094
Random Forest	0.587	0.836	0.572	0.854	0.0963
Extra Trees	0.581	0.826	0.556	0.843	0.1048
XGBoost	0.608	0.823	0.495	0.826	0.0977
LightGBM	0.576	0.823	0.514	0.846	0.0990

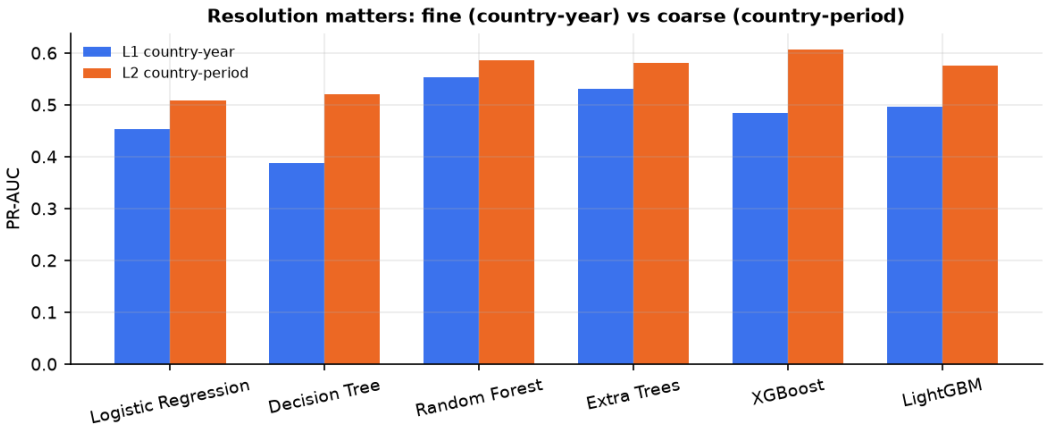


Figure 9: Resolution matters: the same models on fine versus coarse temporal units.

8.4 Ablation settings

Table 12: Ablation settings (AS). AS-7 adds two research-output proxies as a surveillance-bias probe.

Setting	k	PR-AUC (time)	ROC-AUC (time)	PR-AUC (LOCO)
AS-1 history only	7	0.407	0.761	0.174
AS-2 population only	14	0.432	0.850	0.241
AS-3 climate only	10	0.395	0.769	0.078
AS-4 socio-geography	11	0.469	0.824	0.160
AS-5 history+population	21	0.451	0.831	0.321
AS-6 full (all groups)	42	0.487	0.840	0.259
AS-7 full + surveillance proxy	44	0.497	0.831	0.261

Two findings here are more useful than the headline score. (i) *Parsimony wins out-of-country*: AS-5 (21 features) generalises better (0.321 LOCO PR-AUC) than the full 42-feature set (0.259). Adding climate and socio-economic columns buys in-distribution fit and costs transfer. (ii) *Climate alone is nearly useless out-of-country* (0.078 LOCO PR-AUC): it mostly encodes latitude, which is country identity in disguise. The AS-7 probe moves the score by +0.001 LOCO PR-AUC, which is weak evidence that the model is not *purely* tracking research output.

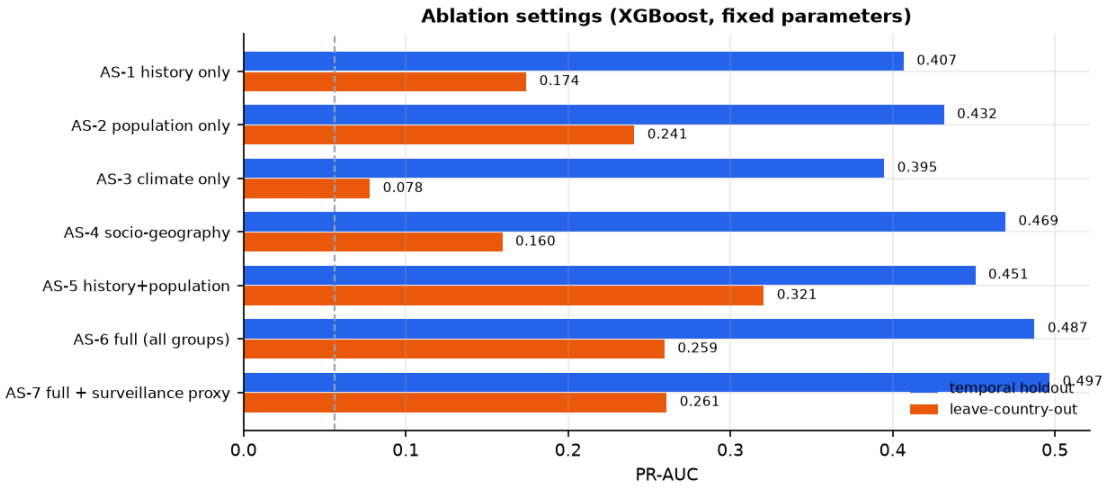


Figure 10: Ablation settings under both schemes.

8.5 The chained pipeline

Stage	Metric	Value
Stage 1 (population forecast)	MAE / median AE	0.424 / 0.174
	Spearman	0.976
Stage 3 (intensity | event)	MAE / median AE	0.150 / 0.079
	Spearman	0.465
Hurdle chain (expected burden)	Spearman	0.326
	events in top 10%	70/106

Table 13: Effect of chaining the population forecast into the occurrence model.

Stage-2 variant	PR-AUC (time)	ROC-AUC (time)	PR-AUC (LOCO)
Plain (no chained feature)	0.530	0.854	0.363
Chained (population forecast added)	0.527	0.853	0.344

Chaining does not improve discrimination. The difference is -0.003 PR-AUC, well inside the bootstrap interval. This is reported as a negative result rather than tuned away: because the population level is already supplied as a direct feature, forecasting it adds no information for the ranking task. The chain earns its place at the *alert* layer, where the hurdle product captures 70 of 106 events in the top decile – not at the discrimination layer.

8.6 Subgroup performance

Table 14: Leave-country-out performance by World Bank region and income stratum.

Stratified by	Group	n	events	ROC-AUC	PR-AUC
region	East Asia & Pacific	456	48	0.865	0.396
region	Europe & Central Asia	912	31	0.826	0.163
region	Latin America & Caribbean	551	12	0.903	0.381
region	Middle East, North Africa, Afghanistan & Pakistan	380	23	0.834	0.220
region	Sub-Saharan Africa	665	27	0.699	0.073
income	High income	1045	66	0.875	0.303
income	Low income	361	10	0.720	0.062
income	Lower middle income	741	42	0.878	0.541
income	Upper middle income	912	54	0.867	0.329

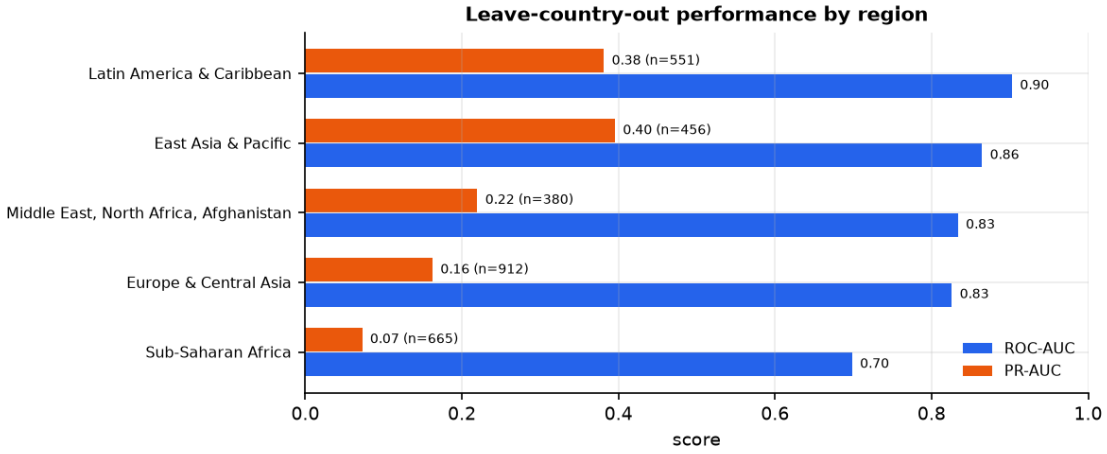


Figure 11: Regional transfer performance under leave-country-out.

This is the study's most policy-relevant table, and it is uncomfortable: performance is lowest in Sub-Saharan Africa and in low-income countries – precisely where glanders surveillance is thinnest

and where an early-warning system would be most valuable. The mechanism is label sparsity, not model failure. We report it rather than averaging it away, because a deployment decision should not treat these strata as equivalent.

8.7 Caveats and limitations

- **The label is documented activity, not incidence.** A zero means "no retrieved record", which conflates absence of disease with absence of publication. All scores should be read as skill at predicting *surveillance-observable* events.
- **Text-mined labels inherit extraction error.** The rule-based gazetteer/lexicon extractor has not been validated against a hand-annotated gold set; false positives are possible where country names appear in author affiliations or general framing.
- **Country-level aggregation hides district heterogeneity.** Glanders in India is heavily concentrated in a few states; a country-year label cannot express that.
- **Small positive count.** With 172 positives, bootstrap intervals are wide (roughly ± 0.11 on the headline PR-AUC) and subgroup estimates are noisy.
- **Associations are not causal.** Geographic and socio-economic features act largely as reporting-capacity proxies; no intervention claim follows from them.

Reproducibility statement

- **Code.** `src/00_novelty_audit.py` through `src/07_make_latex.py`, run in order.
- **Data.** `data/raw` holds the cached upstream payloads; `data/processed` holds the assembled panel and extracted events.
- **Numbers.** `reports/results.json`, `reports/dataset_analysis.json` and `reports/model_comparison.csv`.
- Every figure and table above is regenerated from those files by the scripts; this report contains no hand-transcribed results.

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